Case 1. $A = \mathbb{C}^n / \tau\mathbb{Z}^n \oplus \mathbb{Z}^n$

lattice:

$$\Pi = (\lambda_1 \dots \lambda_{2n}) = \left( \tau \mid Id_n \right)$$

$$(\lambda_1 \dots \lambda_{2n}) \overset{*}{=} (e_1, \dots, e_n)\,\Pi$$

polarization:

$$J_D = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} \overset{*}{=} \left(\omega(\lambda_i, \lambda_j)\right)_{i,j=1}^{2n}$$

where

$$D = \mathrm{diag}(d_1, \dots, d_n) \in M^{n\times n}(\mathbb{Z})$$

$d_\nu \geq 0 \qquad d_\nu \mid d_{\nu+1} \leftarrow$ will be loosed ?

Riemann conditions reduce to

$$\begin{cases} D^{-1}\tau^T = \tau D^{-1} & \Leftrightarrow \quad D\tau = \tau^T D \qquad (RC1) \\ i D^{-1}\tau^H - i\tau D^{-1} > 0 & \qquad (RC2) \end{cases}$$

* You can expand the expression as $\qquad x_\gamma \in \mathbb{C}^n$

$$\begin{aligned} \omega\left(\Pi\begin{pmatrix} x \\ y \end{pmatrix}, \Pi\begin{pmatrix} x' \\ y' \end{pmatrix}\right) &= \omega(x + \tau y, x' + \tau y') \\ &= \omega\left(\sum_i x_i \lambda_i + \sum_i y_i \lambda_{i+n}, \sum_i x_i' \lambda_i + \sum_i y_i' \lambda_{i+n}\right) \\ &= \sum_i d_i x_i y_i' - \sum_i d_i x_i' y_i \\ &= x^T D y' - y^T D x' \\ &= (x^T \; y^T) \begin{pmatrix} & D \\ -D & \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} \end{aligned}$$

CM structure.

$$e_0 (e_1, \dots, e_n) = (e_1, \dots, e_n) \underbrace{\begin{pmatrix} & -d\,1_m \\ 1_m & \end{pmatrix}}_{u}{}_{n\times n} \quad (\mathbb{Q}\text{-coef})$$

If $e_0$ defines a ~~CM~~ K-action, then $e_0$ should fix the lattice, i.e.,

$$e_0 (\tau \mathbb{Z}^n \oplus \mathbb{Z}^n) \subseteq \tau \mathbb{Z}^n \oplus \mathbb{Z}^n$$

we only need that (CM1)

$$e_0 \tau \mathbb{Z}^n \subseteq \tau \mathbb{Z}^n \oplus \mathbb{Z}^n.$$

The best case: (CM2)

$$e_0 \tau = \tau e_0$$

$$u \tau = \tau u \qquad (\text{matrix symbol})$$

If so, then

$$e_0 (\tau \ \ Id) = (\tau \ \ Id) \begin{pmatrix} u & \\ & u \end{pmatrix} \Rightarrow J_u = \begin{pmatrix} & u \\ u & \end{pmatrix}$$

Solving (CM2) yields

$$\tau = \begin{pmatrix} \tau_1 & d\tau_2 \\ -\tau_2 & \tau_1 \end{pmatrix} \qquad (CM2)$$

~~The~~ Riemann condition 1 further ~~also~~ reduces to $D = \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix}$

$$\begin{cases} \tau_1 = \tau_1^T \\ d D_1 \tau_2 = -\tau_2^T D_2 \\ -D_2 \tau_2 = d \tau_2^T D_1 \end{cases}$$

Two special cases solve this equations:

$$\begin{cases} \tau_1^T = \tau_1 \\ \tau_2 \equiv 0 \end{cases} \quad (RC3) \qquad \text{or} \qquad \begin{cases} \tau_1^T = \tau_1 \\ \tau_2^T = -\tau_2 \\ D_2 = d D_1 \\ D_1 \text{ is scalar} \end{cases} \quad (RC4)$$

Compatability

Expand $J_u^T J_D J_u = d J_D$ to

$$\begin{pmatrix} u^T & \\ & u^T \end{pmatrix} \begin{pmatrix} \cancel{0} & D \\ -D & \cancel{0} \end{pmatrix} \begin{pmatrix} u & \\ & u \end{pmatrix} = d \begin{pmatrix} & D \\ -D & \end{pmatrix}$$

$$\shortparallel$$

$$\begin{pmatrix} & u^T D u \\ -u^T D u & \end{pmatrix}$$

Expand $u^T D u = dD$ to

$$\begin{pmatrix} -d & 1 \end{pmatrix} \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix} \begin{pmatrix} 1 & -d \end{pmatrix} = d \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix}$$

$$\shortparallel$$

$$\begin{pmatrix} D_2 & \\ & d^2 D_1 \end{pmatrix}$$

we get the solution

$$D_2 = d D_1 \qquad \text{(Comp)}$$

Hermitian:
~~the He~~

$$H_\lambda = J_D J_u + \sqrt{-d}\, J_D$$

$$= \begin{pmatrix} \cancel{0} & D \\ -D & \end{pmatrix} \begin{pmatrix} u & \\ & u \end{pmatrix} + \sqrt{-d} \begin{pmatrix} & D \\ -D & \end{pmatrix}$$

$$= \begin{pmatrix} \cancel{\sqrt{-d}D} & Du + \sqrt{-d}D \\ -Du - \sqrt{-d}D & \end{pmatrix}$$

$$= \begin{pmatrix} & \sqrt{-d}D_1 & -dD_1 \\ & D_2 & \sqrt{-d}D_2 \\ -\sqrt{-d}D_1 & dD_1 & \\ -D_2 & -\sqrt{-d}D_2 & \end{pmatrix}$$

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$$D\mu + \sqrt{-d}\, D = \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix} \begin{pmatrix} & -d \\ 1 & \end{pmatrix} + \sqrt{-d} \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{-d}\, D_1 & -d\, D_1 \\ D_2 & \sqrt{-d}\, D_2 \end{pmatrix}$$

With the K-basis $\{\lambda_1, \ldots, \lambda_m, \lambda_{n+1}, \ldots, \lambda_{n+m}\}$, the actual Hermitian is

$$H_{K\text{-basis}} = \begin{pmatrix} & \sqrt{-d}\, D_1 \\ -\sqrt{-d}\, D_1 & \end{pmatrix}$$

with

$$\det H_{K\text{-basis}} = (-1)^m d^m d_1^2 \cdots d_m^2$$

$$= (-1)^m \det D$$

$$\equiv (-1)^m$$

$$\begin{tikzcd}
(RC3) \text{ or } (RC4) \arrow[r, Rightarrow, "(CM_2)"] \arrow[d, Rightarrow] & (RC1) \text{ } (RC2) \arrow[d, Rightarrow] \\
(CM_2) \arrow[r, Rightarrow] \arrow[d, Rightarrow] & (CM1) \arrow[l, Rightarrow, "?"] \\
(Comp) &
\end{tikzcd}$$

<u>Conclusion.</u> In the case $\Pi = (\tau \mid Id_n)$

$$J_D = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$$

$$u = \begin{pmatrix} & -d \\ 1 & \end{pmatrix} \rightsquigarrow J_u = \begin{pmatrix} & u \\ u & \end{pmatrix}$$

If $\tau = \begin{pmatrix} \tau_1 & d\tau_2 \\ -\tau_2 & \tau_1 \end{pmatrix}$ satisfy $(CM_2)$ $iD^{-1}\tau^H - i\tau D^{-1} > 0$ $(RC2)$ and

$$\begin{cases} \tau_1^T = \tau_1 \\ \tau_2 \equiv 0 \\ D_2 = dD_1 \end{cases} \quad \text{or} \quad \begin{cases} \tau_1^T = \tau_1 \\ \tau_2^T = -\tau_2 \\ D_2 = dD_1 \\ D_1 \text{ is scalar.} \end{cases}$$

$(RC3) + (Comp)$ $\qquad$ $(RC4)$

then these data defines an AV of Weil type with polarizations

with

$$\det H_{K\text{-basis}} = (-1)^m \cancel{d_1 d_1 \cdots d_m}^{\det D}.$$

$$\equiv (-1)^m$$

Case 2. $A = \mathbb{C}^n / \tau\mathbb{Z}^n \oplus D\mathbb{Z}^n$

lattice: $\Pi = (\lambda_1, \dots, \lambda_{2n}) = \left( \tau \mid D \right)$

polarization: $J_D = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$

Riemann conditions reduce to

$$\begin{cases} \tau^T = \tau & \text{(RC1)} \\ \operatorname{Im}\tau > 0 & \text{(RC2)} \end{cases}$$

CM structure:

$$e_0 (d_1 e_1, \dots, d_n e_n) = (d_1 e_1, \dots, d_n e_n) \begin{pmatrix} & -d 1_m \\ 1_m & \end{pmatrix}_{n \times n}$$

$$\Rightarrow u = D \begin{pmatrix} & -d \\ 1 & \end{pmatrix} D^{-1} = \begin{pmatrix} & -dD_1D_2^{-1} \\ D_2D_1^{-1} & \end{pmatrix} \in M^{n\times n}(\mathbb{Q})$$

not $\mathbb{Z}$!

If

$$e_0 (\tau\mathbb{Z}^n \oplus D\mathbb{Z}^n) \subseteq \tau\mathbb{Z}^n \oplus D\mathbb{Z}^n$$

$$e_0 \tau \mathbb{Z}^n \subseteq \tau\mathbb{Z}^n \oplus D\mathbb{Z}^n \qquad \text{(CM1)}$$

The best case:

$$e_0 \tau = \tau D^{-1} e_0 D$$

$$u \tau = \tau D^{-1} u D \quad \text{(matrix symbol)} \qquad \text{(CM2)}$$

If so, then

$$e_0 (\tau \ D) = (\tau \ D) \begin{pmatrix} D^{-1}uD & \\ & D^{-1}uD \end{pmatrix} \Rightarrow J_u = \begin{pmatrix} D^{-1}uD & \\ & D^{-1}uD \end{pmatrix}$$

Solving (CM2) yields

$$\begin{pmatrix} & -dD_1D_2^{-1} \\ D_2D_1^{-1} & \end{pmatrix}\begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{pmatrix} = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{pmatrix}\begin{pmatrix} & -d \\ 1 & \end{pmatrix}$$

$$\begin{pmatrix} -dD_1D_2^{-1}\tau_{21} & -dD_1D_2^{-1}\tau_{22} \\ D_2D_1^{-1}\tau_{11} & D_2D_1^{-1}\tau_{12} \end{pmatrix} = \begin{pmatrix} \tau_{12} & -d\tau_{11} \\ \tau_{22} & -d\tau_{21} \end{pmatrix}$$

$$\begin{cases} \tau_{22} = D_2D_1^{-1}\tau_{11} \\ \tau_{12} = -dD_1D_2^{-1}\tau_{21} \end{cases}$$

$$\tau = \begin{pmatrix} \tau_1 & -dD_1D_2^{-1}\tau_2 \\ \tau_2 & D_2D_1^{-1}\tau_1 \end{pmatrix} \qquad \text{(CM2)}$$

~~Riemann Condition~~ (RC1) further reduces to

$$\begin{cases} \tau_1^T = \tau_1 \\ \tau_2^T = -dD_1D_2^{-1}\tau_2 \\ (D_2D_1^{-1}\tau_1)^T = D_2D_1^{-1}\tau_1 \end{cases}$$

Compatability

Expand $J_u^T J_D J_u = d J_D$ to $\quad (v = D^{-1} u D = \begin{pmatrix} 1 & -d \end{pmatrix})$

$$\begin{pmatrix} D^{-1} u^T D^{-1} & \\ & D^{-1} u^T D^{-1} \end{pmatrix} \begin{pmatrix} & D \\ -D & \end{pmatrix} \begin{pmatrix} D^{-1} u D & \\ & D^{-1} u D \end{pmatrix} = d \begin{pmatrix} & D \\ -D & \end{pmatrix}$$

$$\shortparallel$$

$$\begin{pmatrix} & v^T D v \\ -v^T D v & \end{pmatrix}$$

Expand $v^T D v = d D$ to

$$\begin{pmatrix} -d & 1 \end{pmatrix} \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix} \begin{pmatrix} 1 & -d \end{pmatrix} = d \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix}$$

$$\shortparallel$$

$$\begin{pmatrix} D_2 & \\ & d^2 D_1 \end{pmatrix}$$

we get the solution

$$D_2 = d D_1 \qquad \text{(Comp)}$$

Under (Comp), (CM2) ~~dec~~ reduces to

$$\tau = \begin{pmatrix} \tau_1 & -\tau_2 \\ \tau_2 & d\tau_1 \end{pmatrix}$$

and (RC1) further reduces to

$$\begin{cases} \tau_1^T = \tau_1 \\ \tau_2^T = -\tau_2 \end{cases}$$

Hermitian

$$H_\lambda = J_D J_u + \sqrt{-d}\, J_D$$

$$= \begin{pmatrix} & D \\ -D & \end{pmatrix} \begin{pmatrix} D^{-1}uD & \\ & D^{-1}uD \end{pmatrix} + \sqrt{-d} \begin{pmatrix} & D \\ -D & \end{pmatrix}$$

$$= \begin{pmatrix} & uD \\ -uD & \end{pmatrix} + \sqrt{-d} \begin{pmatrix} & D \\ -D & \end{pmatrix}$$

$$= \begin{pmatrix} & uD + \sqrt{-d}\, D \\ -uD - \sqrt{-d}\, D & \end{pmatrix}$$

$$= \begin{pmatrix} & \begin{matrix} \sqrt{-d} D_1 & -d D_2 \\ D_2 & \sqrt{-d} D_2 \end{matrix} \\ \begin{matrix} -\sqrt{-d} D_1 & d D_2 \\ -D_2 & -\sqrt{-d} D_2 \end{matrix} & \end{pmatrix}$$

$\cancel{\text{tr}}$ $\cancel{uD}$

$\cancel{\text{tr}}$ $$uD = \begin{pmatrix} & -d D_1 D_2^{-1} \\ D_2 D_1^{-1} & \end{pmatrix} \begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix} = \begin{pmatrix} & -d D_1 \\ D_2 & \end{pmatrix}$$

With the K-basis $\{\lambda_1, \dots, \lambda_m, \lambda_{n+1}, \dots, \lambda_{n+m}\}$, the actual Hermitian is

$$H_{K\text{-basis}} = \begin{pmatrix} & \sqrt{-d}\, D_1 \\ -\sqrt{-d}\, D_1 & \end{pmatrix}$$

with

$$\det H_{K\text{-basis}} = (-1)^m d^m d_1^2 \cdots d_m^2$$

$$= (-1)^m \det D$$

<u>Conclusion</u>. In the case $\Pi = (\tau \mid D)$

$$J_D = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$$

$$u = \begin{pmatrix} & -dD_1D_2^{-1} \\ D_2D_1^{-1} & \end{pmatrix} \rightsquigarrow J_u = \begin{pmatrix} v & \\ & v \end{pmatrix}$$

If $\tau = \begin{pmatrix} \tau_1 & -\tau_2 \\ \tau_2 & d\tau_1 \end{pmatrix}$ satisfy $\cancel{(RC1)+(RC2)+}$

$$\begin{cases} \tau_1^T = \tau_1 & (RC1) \\ \tau_2^T = -\tau_2 & \\ \operatorname{Im}\tau > 0 & (RC2) \\ D_2 = dD_1 & (\text{Comp}) \end{cases}$$

then these data defines an AV of Weil type with polarizations,

with

$$\det H_{K\text{-basis}} = (-1)^m \overset{\det D}{\cancel{d^m d_1^2 \cdots d_m^2}}.$$

Rmk. Even if sometimes (Comp) is not satisfied, one can produce a new polarization

$$d\omega + e_0^*\omega = dJ_D + J_u^T J_D J_u := J_{D'}$$

which satisfies (Comp).

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$$J_u^T J_{D'} J_u = dJ_u^T J_D J_u + (J_u^T)^2 J_D J_u^2 = dJ_{D'}$$

In case 1,

$$J_{D'} = dJ_D + J_u^T J_D J_u$$

$$= d\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} + \begin{pmatrix} u^T & \\ & u^T \end{pmatrix}\begin{pmatrix} & D \\ -D & \end{pmatrix}\begin{pmatrix} u & \\ & u \end{pmatrix}$$

$$= \begin{pmatrix} 0 & dD + u^T D u \\ -dD - u^T D u & 0 \end{pmatrix}$$

$$D' = dD + u^T D u$$

$$= d\begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix} + \begin{pmatrix} & 1 \\ -d & \end{pmatrix}\begin{pmatrix} D_1 & \\ & D_2 \end{pmatrix}\begin{pmatrix} & -d \\ 1 & \end{pmatrix}$$

$$= \begin{pmatrix} dD_1 & \\ & dD_2 \end{pmatrix} + \begin{pmatrix} D_2 & \\ & d^2 D_1 \end{pmatrix}$$

$$= \begin{pmatrix} dD_1 + D_2 & \\ & d(dD_1 + D_2) \end{pmatrix} \triangleq \begin{pmatrix} D_1' & \\ & D_2' \end{pmatrix}$$

$$H_\lambda = J_{D'} J_u + \sqrt{-d}\, J_{D'}$$

$$H_{k\text{-basis}} = \begin{pmatrix} & \sqrt{-d}\, D_1' \\ -\sqrt{-d}\, D_1' & \end{pmatrix}$$

$$\det H_{k\text{-basis}} = (-1)^m \det D'$$

In case 2,

$$J_{D'} = d J_D + J_u^T J_D J_u$$

$$= \begin{pmatrix} 0 & dD + v^T D v \\ -dD - v^T D v & 0 \end{pmatrix}$$

$$D' = dD + v^T D v$$

$$= \begin{pmatrix} dD_1 + D_2 & \\ & d(dD_1 + D_2) \end{pmatrix} \triangleq \begin{pmatrix} D_1' & \\ & D_2' \end{pmatrix}$$

$$H_\lambda = J_{D'} J_u + \sqrt{-d}\, J_{D'}$$

$$H_{k\text{-basis}} = \begin{pmatrix} & \sqrt{-d}\, D_1' \\ -\sqrt{-d}\, D_1' & \end{pmatrix}$$

$$\det H_{k\text{-basis}} = (-1)^m \det D'$$